

VAIBHAV PANCHAL

AI/ML Developer | Deep Learning Researcher | AI Full-Stack Engineer

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PROFESSIONAL SUMMARY

AI/ML Developer at TestGrid Labs Inc. building intelligent test automation pipelines using deep learning and LLMs. M.E. in Software Engineering (LDCE, GTU, CGPA 8.21). Research published in Computer Vision (YOLOv11 + ECA, +3.8% mAP50 on DIOR-R) and Cybersecurity (hybrid IDS, 98.4% accuracy). Hands-on across the full ML lifecycle from data engineering and model training to FastAPI deployment and CI/CD integration. Core stack: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LangChain, Docker.

TECHNICAL SKILLS

Languages	Python (Advanced), SQL, JavaScript, C/C++, HTML/CSS, java
ML / DL	PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, XGBoost
AI / Research	Computer Vision, NLP, LLMs, RAG, YOLOv11-OBb, ECA Attention, Object Detection
MLOps & Tools	Docker, Git, FastAPI, Streamlit, Jupyter, MLflow, Flask, Postman
Automation & Tools	Selenium Grid, Git, Bitbucket
Web	HTML, CSS, JavaScript, REST APIs

WORK EXPERIENCE

Junior AI/ML Developer | **TestGrid Labs Inc.** | Navsari, Gujarat May 2026 – Present

- Built deep learning inference pipelines for automated UI regression detection, reducing manual test triage time by flagging anomalies before human review.
- Designed and deployed FastAPI-based model-serving endpoints integrated into the CI/CD pipeline, enabling real-time ML inference on test runs at scale.
- Prototyped an LLM-assisted test case generation module using LangChain + Ollama; reduced repetitive test authoring effort across the QA team.

Trainee Associate Software Developer | **Techreale Pvt. Ltd.** | Valsad, Gujarat Jan 2024 – Apr 2024

- Delivered production features for the Famyll healthcare platform (React.js, Node.js, MongoDB) across claim management, auth flows, and real-time communication modules.
- Reduced API response handling bugs by refactoring async integration patterns; contributed across 3 sprint cycles in an Agile team of 8.

React Developer Intern | **INFOLABZ IT Services Pvt. Ltd.** | Ahmedabad, Gujarat Jul 2023 – Aug 2023

- Built and optimised React.js dashboard components with async API integration; improved rendering performance through targeted component-level refactoring.

PROJECTS

Multi-Agent Customer Support System — LangGraph Orchestration

2026

Python · LangGraph · LangChain · FastAPI · Redis · Langfuse

- Built a multi-agent customer support platform with a Supervisor Agent that classifies query intent and routes to specialized Billing, Technical Support, and Refund sub-agents implemented as LangGraph state-graph nodes.
- Preserved context across agent hand-offs via summarized state transfer and persisted conversation memory (Redis + SQLite), so a customer's session survives across requests without repeating themselves.
- Instrumented every agent call, tool call, and hand-off with Langfuse tracing for full observability into how a single ticket moved through the system.

Production RAG System with Evaluation

2026

Python · LangChain · ChromaDB/Pinecone · RAGAS · Streamlit

- Built a RAG pipeline with hybrid retrieval (dense vector similarity + BM25 keyword search) and a citation-enforced grounding prompt to reduce hallucination risk.
- Built a first-class evaluation suite using RAGAS metrics Faithfulness, Context Precision, Answer Relevancy benchmarking every prompt or retrieval change against a held-out Q&A test set before it's considered shipped.
- Tracked evaluation results over time with Langfuse to catch quality regressions before they reached users, surfaced via a live Streamlit metrics dashboard alongside the chat interface.

Intrusion Detection System — CNN + BiLSTM + MultiHead Attention

Apr – May 2025

Python · TensorFlow/Keras · Scikit-learn · PCA · AutoEncoder

- Designed a hybrid CNN–BiLSTM–Attention architecture for multi-class network traffic classification; reached 98.4%+ accuracy on NSL-KDD and CICIDS benchmarks, outperforming Random Forest and SVM baselines.
- Applied PCA and a stacked AutoEncoder for feature compression, reducing input dimensionality by ~60% without accuracy loss improving inference speed for real-time deployment scenarios.
- Structured the codebase as a modular pipeline (ingest → preprocess → train → evaluate) to support easy retraining on new traffic datasets.

ExamForge — LLM-Powered Exam Question Generator

2026

Python · LangChain · Ollama · OpenAI GPT · Streamlit

- Built an end-to-end RAG pipeline: ingests syllabus PDFs, chunks and embeds content, then prompts an LLM to generate structured question papers with configurable difficulty and Bloom's taxonomy tagging.
- Supported both local inference (Ollama) and cloud (OpenAI API), making it deployable in air-gapped institutional environments; cut manual question-drafting time by ~80% in pilot use.
- Engineered structured JSON output parsing with validation to ensure generated questions conform to a strict schema before rendering to PDF.

Content-Based Movie Recommendation Engine

Jan – Feb 2025

Python · Scikit-learn · TF-IDF · Cosine Similarity · Streamlit

- Implemented TF-IDF vectorisation and cosine similarity on the TMDB 5000 dataset; deployed as a real-time Streamlit app with sub-second recommendation latency.
- Pipeline is architected for extensibility collaborative filtering and hybrid approaches can be plugged in without restructuring the serving layer.

RESEARCH & PUBLICATIONS

Channel Attention for Oriented Object Detection in Remote Sensing

2025

YOLOv11-ORB · ECA Attention · DIOR-R Dataset · PyTorch

- Integrated Efficient Channel Attention (ECA) into YOLOv11s-ORB to improve feature recalibration for small and densely packed objects in aerial imagery.

- Evaluated on DIOR-R (20-class remote sensing benchmark): achieved +3.8% mAP50 and +4.3% recall over the baseline, with no increase in parameter count.

Hybrid Deep Learning Architecture for Network Intrusion Detection

2025

CNN + BiLSTM + Attention · NSL-KDD · CICIDS · TensorFlow

- Proposed a sequential CNN–BiLSTM–Attention IDS that captures both spatial and temporal patterns in network flow data; applied PCA and AutoEncoder for feature compression.
- Achieved 98.4%+ accuracy on NSL-KDD and CICIDS; outperformed Random Forest and SVM across all attack categories on precision, recall, and F1-score.

EDUCATION

Degree / Exam	University	Institute	Year	Score
M.E. – Software Engineering	GTU	LDCE Ahmedabad	2024–2026	CGPA 8.21
B.E. – Computer Engineering	GTU	GDEC Navsari	2020–2024	CGPA 8.11

ACHIEVEMENTS & CERTIFICATIONS

- SSIP Hackathon 2022 — Developed an automated smart attendance management system; selected among top teams from Gujarat.
- Dev-o-lution Hackathon — Certificate of Participation for building a competitive solution under a 24-hour constraint.
- 2 research papers in Computer Vision (remote sensing, +3.8% mAP50) and Cybersecurity (IDS, 98.4% accuracy).

INTERESTS

AI/ML Research, LLMs & RAG Systems, Computer Vision, Deep Learning, MLOps, Automation, Cloud Computing, Data Science | Chess, Exploring Emerging Technologies, Music, Video Games